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REVIEW PAPER

Mycelium-Based Breakthroughs: Exploring Commercialization, Research, and Next-Gen Possibilities

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Abstract
Mycelium-based composites (MBCs) and pure-mycelium materials (PMMs) represent groundbreaking advancements in materials engineering, offering environmentally sustainable alternatives aligned with the Sustainable Development Goals (SDGs). These innovative materials integrate mycelium, a natural biological adhesive, with lignocellulosic by-products, demonstrating a multidisciplinary approach that merges biology, engineering, and design to address critical environmental challenges. This comprehensive review explores recent advancements in MBC technology, encompassing commercial applications, ongoing research and development, scientific literature, mycological insights, patents, and intellectual property. Selected product samples illustrate the versatility and potential of MBCs. The pivotal role of mycelium in addressing current and future challenges in materials science is underscored. The review identifies eleven distinct application areas for MBCs and PMMs, including five commercialized sectors and six under active research and development. Furthermore, key directions for future studies are outlined, emphasizing the integration of MBCs and PMMs to enhance the sustainability of conventional materials and drive the evolution of next-generation materials.

Keywords Mycelium · Environmental Sustainability · Mycelium-based Composites · Lignocellulosic by-products · Next-gen Materials · Bio-based Materials

Introduction

Amid escalating environmental and sustainability crises, global dependence on synthetic materials, particularly plastics, presents a critical challenge. Plastic production has surged from approximately 1.5 million tons per year in 1950 to an estimated 359 million tons by 2018, significantly contributing to environmental degradation [1–3].

The production of synthetic materials is energy-intensive and generates emissions detrimental to both the environment and human health. Plastics pose a severe ecological threat

Extended author information available on the last page of the article

due to their prolonged degradation timelines and persistent environmental contamination [4–6]. The growing problem of fossil-fuel-based pollution and plastic waste has far-reaching consequences for biodiversity, energy efficiency, and economic sustainability, necessitating an urgent transition toward biodegradable alternatives that harness the inherent sustainability of natural materials [7, 8].

In response to these challenges, mycelium-based composites (MBCs) and pure-mycelium materials (PMMs) have emerged as promising eco-friendly alternatives [9–12]. These materials are produced by combining fungal mycelia—with its natural binding properties—with lignocellulosic biomass derived from agricultural by-products such as sugarcane bagasse, spent coffee grounds, and rice husks [13–16]. Lignocellulosic by-products, rich in cellulose, hemicellulose, and lignin, serve as essential substrates, providing nutrients for mycelial growth while supporting the formation of dense, interwoven hyphal networks that contribute to the mechanical integrity of the final composite. The physicochemical properties of these substrates, including particle size, porosity, and surface roughness, directly influence mycelial colonization, binding efficiency, and the resulting mechanical properties, such as compressive strength, elasticity, and water absorption [14, 15]. MBCs fabricated by different lignocellulosic waste, for example hemp, flax, flax waste, soft wood, and wheat straw, showed different content of cellulose, hemicellulose, and lignin as well as different substrate particle size [13]. Therefore, it led to variety of density, compressive modulus, and shrinkage. While chopped flax resulted in the highest compressive modulus, a loose pine wood gave the lowest one. Moreover, pineapple fiber has been proved to improve more than 2-fold of flexural strength for MBC from spent coffee ground [16].

Furthermore, recent advancements in substrate pre-treatment techniques, such as steam explosion, alkali treatment, and enzymatic hydrolysis, have been shown to enhance substrate compatibility, improve bioavailability, and increase mycelial binding efficiency. By utilizing these agricultural residues, the production of MBCs and PMMs not only offers an eco-friendly alternative to synthetic materials but also facilitates a sustainable method for valorizing agricultural waste into materials with both low and high economic value [17–19]. This approach highlights the critical role of lignocellulosic by-products in shaping the properties and applications of mycelium-based materials, while contributing to circular economic systems and reducing environmental impact.

Characterized by reduced cost and density, alongside key attributes such as biodegradability, lightweight composition, fire and water resistance, economic feasibility, local sourcing, rapid growth, and energy efficiency, MBCs present a versatile alternative to conventional synthetic and construction materials [18, 20–22]. Their successful application in construction, packaging, and furniture production can help mitigate critical environmental issues, including plastic pollution and the excessive consumption of natural resources in construction [23–26].

This review aims to consolidate and critically assess current research and emerging concepts on utilizing agricultural waste and fungal species for the synthesis of MBCs and PMMs. Fungi play a crucial role in advancing sustainable material technologies, paving the way for innovations in materials science and industry [27–29]. We explore the impact of innovative designs and patent trends in mycelium-based materials (MBMs), as well as commercial products and prototypes, in facilitating the transition from a petroleum-based economy to a bio-based, circular, and green economy aligned with the Sustainable Development Goals (SDGs) [30–34].

MBC and PMM applications are categorized into two main groups: commercialized products and those in the research and development phase. Commercialized applications include construction materials, artistic and product design, 3D printing, furniture, packaging, textiles, fabrics, paper-like materials, fashion, and applications in the food and beverage industry. Meanwhile, ongoing research focuses on environmental remediation, agriculture and horticulture, health and medical devices, amulets and religious artifacts, space and automotive industries, electronic devices, and energy production.

Concept and Development of mycelium-Based Products

Fungi exhibit a remarkable ability to convert organic matter into a wide range of valuable products, offering innovative solutions to pressing global challenges [35]. Fungal biotechnology plays a pivotal role in transitioning from a petroleum-dependent economy to a bio-based circular economy, enabling the sustainable production of resilient resources across multiple sectors, including food, feed, chemicals, fuels, textiles, and materials for construction, automotive, transportation, and furniture industries [36–39].

Since the late 20th century, advancements in biotechnology and a growing emphasis on environmental sustainability have driven research into mycelium-based applications. These applications fall into two material categories: mycelium-based composites (MBCs) and pure-mycelium materials (PMMs) (Fig. 1A). MBCs are created by integrating mycelium with organic substrates. Early research focused on leveraging mycelium for composite materials by inoculating fungal spores into organic substrates.

As illustrated in Fig. 1B, MBC production begins with selecting an agricultural waste substrate and fungal species. The substrate is sterilized to prevent contamination before culturing the fungus. The mycelium then digests the substrate and binds it into a cohesive solid mass through a process known as myceliation, yielding materials that are lightweight, strong, and naturally biodegradable [21, 40].

Following cultivation, post-processing is carried out to deactivate fungal growth and enhance the material's properties. Various substrates, including rice bran, sawdust, and corn pericarp, have been explored in combination with mycelium to tailor material characteristics [41, 42]. Beyond digestible substrates, non-digestible materials such as textile waste, sand, and clay have also been utilized. Given that the substrate constitutes 70–80 wt% of MBC, its properties are influenced not only by the fungal species but also by factors such as substrate composition, moisture content, temperature, nutrient availability, pH, and oxygen levels. The interaction between fungal species and substrate types yields diverse MBC properties, mimicking natural materials, foam-like substances, and polymers [43, 44]. Modern engineering applications of MBCs have significantly expanded, encompassing eco-friendly packaging solutions and construction materials such as bricks and insulation panels.

Unlike MBCs, whose properties are highly dependent on substrate type, pure-mycelium materials (PMMs) are fabricated exclusively from mycelium fibers or biomass, typically forming a mat or layer [45] (Fig. 1A). As a result, PMM properties are influenced solely by the fungal species, with different species producing distinct mycelium arrangements and compositions. Mycelium naturally grows into a network-like microstructure, where fibers interweave at the microscale [46]. This micro-woven characteristic enhances PMM strength. The mycelium cell wall comprises polysaccharides, chitin/chitosan, and glucan

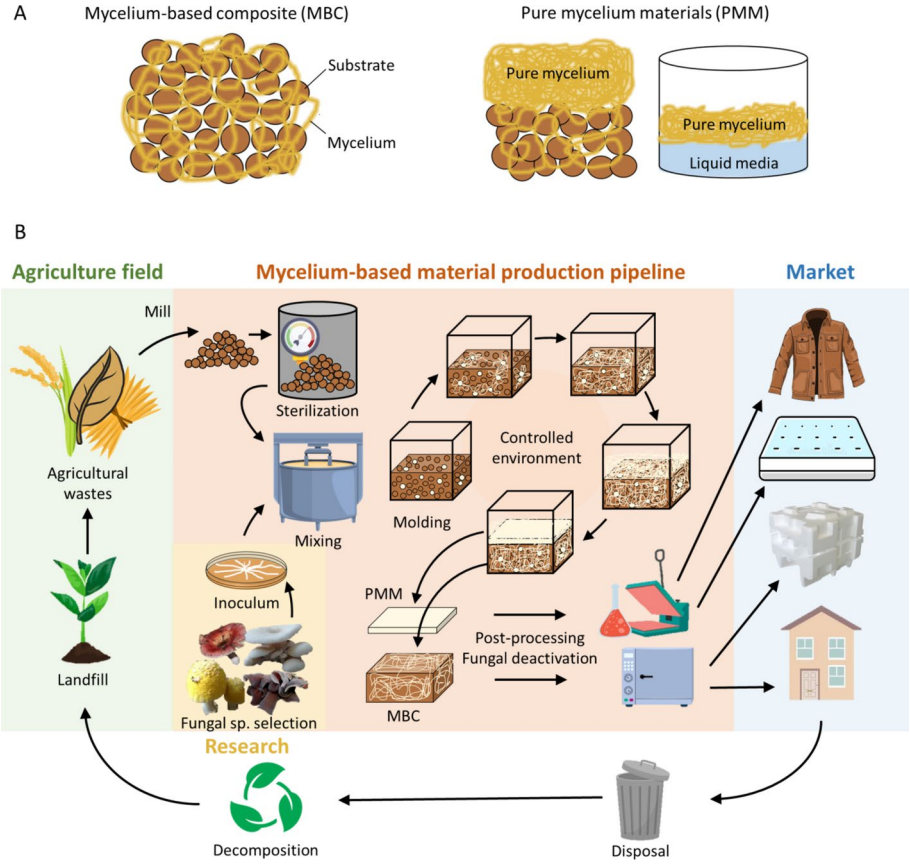

Fig. 1 Concept of MBMs, encompassing MBC and PMM (**A**). Diagram illustrating the production workflow of the myco-fabrication process, detailing each stage from the agricultural sector to the market (**B**).

biopolymers, with the ratio of chitin/chitosan to glucan dictating its mechanical properties [33].

Variations in fungal species and growth conditions influence PMM growth patterns and polysaccharide composition. Additionally, PMMs exhibit properties characteristic of both polymers and elastomers and can be chemically or physically modified for diverse applications. Notably, PMMs are utilized as leather alternatives and textiles, demonstrating their versatility and potential in sustainable fashion and beyond [47, 48].

Advancements in mycelium-based materials (MBMs) demonstrate their ability to replace polymers, elastomers, and natural materials, positioning them as next-generation alternatives [18, 20, 49]. Mycelium technology continues to evolve, propelled by ongoing research and innovation. Scientists and engineers are exploring novel substrates, optimizing growth conditions, and refining fabrication techniques to enhance the mechanical properties and functionality of MBCs. This progression—from traditional applications to contemporary engineering solutions—underscores the transformative potential of mycelium, paving the way for sustainable, fungi-derived materials to play a pivotal role across various industries [50].

Trend Analysis of MBCs

In Conjunction With Global Development Goals

The concept of a Bio-Circular-Green (BCG) economy offers a transformative approach to addressing global challenges such as environmental degradation, resource scarcity, and climate change. Central to this model is the integration of biological processes and materials into the economic system to create sustainable, circular, and green solutions [51, 52].

Fungi play a vital role in this paradigm by providing innovative and environmentally friendly alternatives applicable across various industries, including construction, packaging, and product design [53–55]. This approach underscores the commitment to sustainable innovation, aligning with the BCG economy's focus on biological processes and green growth [56].

As shown in Fig. 2, myco-composites, as part of this initiative, directly contribute to several SDGs, such as promoting sustainable industrialization (SDG 9), fostering eco-friendly urban development (SDG 11), encouraging responsible consumption (SDG 12), mitigating climate change (SDG 13), and supporting biodiversity (SDG 15) [30, 57, 58].

This integration embodies the essence of sustainability and highlights the significant potential of myco-composites in driving forward the BCG economy. The sustainable production of mycelium biocomposites, which avoid harmful chemicals and processes, supports biodiversity preservation and promotes sustainable land use practices, thereby contributing to ecosystem health.

Moreover, these initiatives align with Environmental, Social, and Governance (ESG) criteria by reducing environmental impact, enhancing community well-being, and promoting responsible governance through sustainable practices. The integration of BCG principles, SDGs, and ESG frameworks within mycelium material technologies exemplifies a holistic

Fig. 2 Diagram depicting the potential applications of fungi in biotechnology and their alignment with SDGs, specifically SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), and SDG 15 (Life on Land)

sustainability model, demonstrating how bio-based innovations foster environmental stewardship, social responsibility, and ethical governance, thereby contributing to a more sustainable and equitable world [31, 59].

The utilization of fungi for myco-composites underscores the interdisciplinary nature of the BCG economy, effectively bridging biology, materials science, and environmental engineering. This cross-disciplinary approach is essential for developing innovative solutions to address the complex challenges of sustainability [18, 60].

Academic and Investment Sectors

A literature survey of mycelium-based materials was conducted in June 2024 with Google Scholar using the keywords: "mycelium-based", "mycelium materials", "fungal materials", "mushroom materials", "mushroom composite", "mycelium composite", "mycelium brick", "mycelium leather", "mycelium packaging", "mycelium foam", "mycelium mat", "mycelium sheet", "mycelium insulation", "mycelium textile", "mycelium absorbent", "mycelium design", "mycelium construction", "mycelium furniture", and "mycelium meat". Patent analysis was conducted using Patent Analytics by Patsnap with similar keywords. The search period spanned from 2011 to 2024, and all results were manually screened for relevance to mycelium-based materials. The analysis revealed a significant rise in published articles on mycelium-based materials over the last decade, particularly from 2016 to 2023, where the number of publications increased tenfold (Fig. 3A). Publications on MBMs have recently outpaced those on other fungal products like enzymes, antibiotics, and active compounds. By 2023, more than half of the publications in our literature survey were related to MBM. Ecovative filed a key patent for producing materials and composites using filamentous fungi growth in 2007 (IP no. US9485917B2). This demonstrated the commercial viability of mycelium-based materials, leading to further patents from companies like Mycoworks Inc. and Bolt Threads for mycelium leather manufacturing. Major contributors to the patent landscape include Mycoworks Inc. (60 patents), Bolt Threads (49 patents), Ecovative Design LLC (25 patents), and Biohm Ltd. (10 patents) (Fig. 3B). Notable patent holders include Philip Ross (40 patents), Nicholas Wenner (33 patents), and Jordan Chase

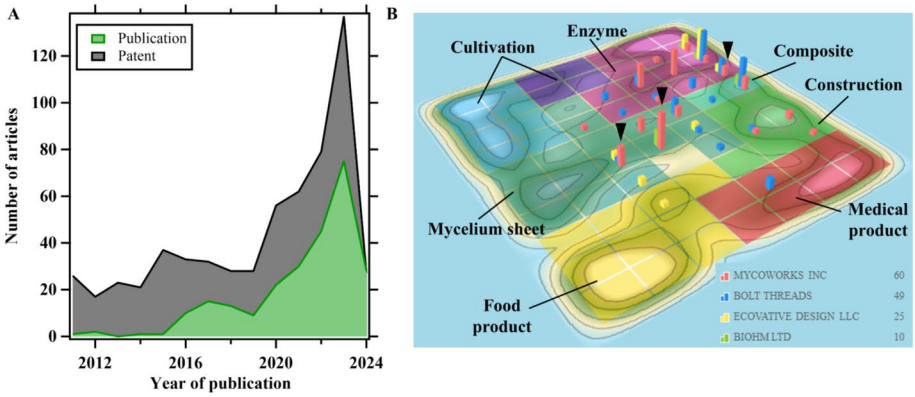

Fig. 3 Publication and patent trends for MBMs (**A**) and patent landscape analysis of existing patents related to MBMs (**B**)

(30 patents). Recent patents predominantly focus on cultivation methods, mechanical property enhancement, and the development of specific mycelium-based products [61].

The patent landscape analysis in Fig. 3B illustrates the range of applications patented by various companies, divided into eight sectors. Most of these applications focus on extracting and cultivating mycelium for its active ingredients and enzymes, primarily used in the food and pharmaceutical industries. MBMs are targeted at five key applications: construction, composites, packaging, pure mushroom sheets, and food applications [49]. For construction materials, MBCs have been developed for building panels, sandwich structures, and architectural models. These composites are valued for their aesthetic qualities and require enhancements in mechanical properties. In packaging, MBC is prized for its lightweight and degradable characteristics. PMMs are mainly fabricated as sheets and foams, with significant potential in food applications as meat alternatives. Various cultivation techniques for mycelium have also been patented [53]. The patent landscape reveals substantial opportunities for further development of mycelium-based materials, which will be explored in the subsequent section of this review.

Current Applications and Potential Ideas of MBCs and their Application

This section offers a comprehensive overview of the current applications of MBCs, categorized into two primary groups. The first group consists of applications with commercialized products that are already available on the market (Sect. 4.1). The second group encompasses applications still under development (Sect. 4.2).

Mycelium-Based Materials Applications With Commercialized Products

The utilization of mycelium as a foundational material in emerging biomaterials represents a rapidly expanding sector within materials innovation. Companies worldwide are leveraging mycelium to develop sustainable alternatives to conventional products. Notably, Ecovative, MycoWorks, and Mogu have explored mycelium's unique properties to create diverse biomaterials, as outlined in Table 1.

The development and application of MBMs indicate a broader shift towards sustainability in material science. Ongoing research and development efforts are focused on enhancing the performance, scalability, and range of applications of MBM, promising further to cement their role in the future of sustainable materials [12, 13, 27]. MBMs and their products are used in various innovative ways across different industries, reflecting the material's versatility and sustainability [62]. Notable examples listed in Table 1 are described below.

Construction, Building Materials/ Infrastructure and Urban Planning/ Architecture Form

Mycelium bricks (myco-bricks) offer a sustainable alternative to traditional construction materials, as evidenced by their Life Cycle Assessment (LCA) [64–66]. Projects like "Hy-Fi" have demonstrated the potential of these materials [67]. Over the past decade, numerous large pavilion projects have highlighted the utility of myco-materials in construction, necessitating a distinction between their use in structural roles and as cladding or surface applications. For example, the "Shell Mycelium" pavilion in India [68], the "Living Pavil-

Table 1 Selected list of mycelium-based products, companies/startups, country of origin, and website/data accessibility in 2024 [62, 63]

Company and (Product name™/ Commercial Product)	Biomimicry	Founders/ owner and (year founded)	Country/ Headquarter	Website/social network availability accessed
NEFFA® | New Fashion Factory (formerly known as MycoTEX®)	Leather/exotic skins and paper-like	Aniela Hoitnk and Nicoline van Enter (2004)	Netherlands	https://www.mycotex.nl/accessed on March 07, 2024
Bolt Threads (Mylo™, B-Silk™ protein and Microsilk™)	Synthetic textiles/ leather and beauty product ingredients	Dan Widmaier and David Breslauer (2009)	USA	https://boltthreads.com/accessed on March 07, 2024
Mycotechnology (ClearIQ™ Natural Flavor)	Food ingredients	Pete Lubar and James Langan (2013)	USA	https://www.mycoiq.com/accessed on March 07, 2024
MycoWorks (Reishi™ and Fine Mycelium™)	Leather/ exotic skins	Philip Ross, Sophia Wang and Eddie Pavlu (2013)	USA	https://www.mycoworks.com/accessed on March 07, 2024
Ecovative Design: MycoFlex, Forager, Hides, AirMycelium, MycoComposite, MycoBoard	Leather/foam, packaging, leather, and exotic skins	Gavin McIntyre, Eben Bayer (2007)	USA	https://www.ecovative.com/https://forager.bio/ (accessed on March 07, 2024)
Mycotech Lab (Mylea™, MYCL™ Merch and MYCL™ Composite)	Leather and board	Annisa Wibi, Adi Reza Nugroho, Ronaldiaz Hartantyo, Arekha Bentangan and Robby Zidna Ilman (2015)	Indonesia	https://mycl.bio/accessed on March 07, 2024
Mogu	Acoustic panels, wall panel and flooring	Maurizio Montalti (2015)	Italy	https://mogu.bio/accessed on March 07, 2024
Polybion (Polybion™)	Biomaterials	Axel Gómez Ortigoza, Alexis Gómez Ortigoza and Bárbara González Rolón (2015)	Mexico and Spain	https://www.polybion.bio/accessed on March 07, 2024
Biohm	Packaging solutions and building Products	Ehab Sayed (2016)	United Kingdom	https://www.biohm.co.uk/accessed on March 07, 2024
Grown.bio	Protective packaging	Jan Berbee and Arthur Moree (2016)	Netherlands	https://www.grown.bio/accessed on March 07, 2024
Biomyc	Packaging	Atanas Enev (2017)	Bulgaria	https://biomyc.eu/accessed on March 07, 2024
Infinite Roots® (formerly Mushlabs)	Food products	Mazen Rizk, Anne-Cathrine Hutz and Thibault Godard (2017)	Germany	https://www.infiniteroots.com/accessed on March 07, 2024

Table 1 (continued)

Company and (Product name™/ Commercial Product)	Biomimicry	Founders/ owner and (year founded)	Country/ Headquarter	Website/social network availability accessed
Spora Biotech (Sporatex)	Leather/ exotic skins	Hernán Rebolledo, José Miguel Figueroa and Leopoldo Naranjo (2017)	Chile	https://spora.co.uk/ accessed on March 07, 2024
Biofab (Mushroom™ Packaging)	Packaging	James Ferrier, Phil Matheson, Michael Khuwattanasenee (2018)	New Zealand	https://www.biofab.bio/accessed on March 07, 2024
Mycelium materials Europe (MyFoam™)	Biofoam and packaging	Bert Rademakers (2018)	Netherlands	www.myceliummaterials.nl/accessed on March 07, 2024
Mycocycle (MycoFILL©, MycoFIBER© and MycoFOAM©)	Down low-carbon biobased materials, waste treatment and disposal	Joanne Rodriguez (2018)	USA	https://mycocycle.com/ accessed on March 07, 2024)
Mylium	Leather/exotic skins	Iris Houthoff (2018)	Netherlands	https://www.mylium.nl/accessed on March 07, 2024
Okom wrks labs (zerø-frm™)	Construction materials	Joshua English, Chris Magwood, John Kabanda, April Magill (2018)	USA	www.okomwrks.co/ accessed on March 07, 2024
Magical mushroom Company (Mushroom® Packaging)	Packaging	Paul Gilligan (2019)	United Kingdom	https://magicalmushroom.com/accessed on March 07, 2024
Michroma.co (Red+colorant)	Food ingredients	Mauricio Braia and Ricardo Cassini (2019)	Argentina and USA	https://www.michroma.co/home/ accessed on March 07, 2024
Mush®	Decoration, fashion, packaging, construction and furniture	Eduar Sydney, Leandro Oshiro and Antonio De Francisco (2019)	Brazil	https://mush.eco/ accessed on March 07, 2024
Mycellium	Plant pot, acoustic panel, and insulation panel	Mahmoud Fawzy, Omer Fathy, Mohammed Elkholy and Mohamed Thrwat (2019)	Egypt	www.mycellium.co/ accessed on March 07, 2024
Mykkö	Fabricated materials	Aurélie fontan and Ashley granter (2019)	United Kingdom	https://www.mykko.co.uk/accessed on March 07, 2024
Bewilder	Foods, decorations and accessories	Ng Sze Kiat (2020)	Singapore	https://www.bewildersg.com/accessed on March 07, 2024
Close the Loop (Loop Living Cocoon™, Loop EarthRise™ and Loop ForestBed™)	Coffins and carriers	Bob Hendrikx and Lonneke Westhoff (2020)	Netherlands	https://loop-biotech.com/accessed on March 07, 2024

Table 1 (continued)

Company and (Product name™/ Commercial Product)	Biomimicry	Founders/ owner and (year founded)	Country/ Headquarter	Website/social network availability accessed
Mycela	Materials	Renée Isabel Jung, Nina Havermans and Sondre Eng (2020)	Norway	https://mycela.no/ accessed on March 07, 2024
Mycel Project (Myco Leather)	Leather substitute	Sungjin Sah, Sungwon Kim and Yunggon Park (2020)	South Korea	https://mycelproject.com/platform/platform_en.php/accessed on March 07, 2024
MycoFutures	Leather/exotic skins	Stephanie Lipp and Leo Gillis (2020)	Canada	https://www.myceliumofthefuture.com/accessed on March 07, 2024
Mushroom material	Packaging	Shaun Seaman (2020)	New Zealand	www.mushroommaterial.com/accessed on March 07, 2024
Nomy	Food and feed products	Ingrid Dynna and David Andrew Quist (2020)	Norway	www.nomy.no/ accessed on March 07, 2024
Comu Labs	Furnitures	Ilia Litviak and Ilya Gindin (2021)	USA	https://comulabs.com/accessed on March 07, 2024
Myceen	acoustic panel, lamps, chairs and tables	Siim Karro and Karl Pärtel (2021)	Estonia	https://myceen.com/ accessed on March 07, 2024
MYCO (MYCO 4Pack, MYCO FixPack, MYCO Go and Comfy and MYCO Fence)	Packaging, shaped fixation, aroma diffusers and carriers of odor concentrates	David Šohaj Minařík (2021)	Czech Republic	https://www.myco.cz/en/accessed on March 07, 2024
Mykor (MykoFoam)	Insulation panel	Olivia Page and Valentina Dipietro (2021)	United Kingdom	https://www.mykor.co.uk/(accessed on March 07, 2024)
Really Clever	Sustainable materials and clean fuels	Patrick Pinto and Matthew Millar (2021)	United Kingdom	https://reallyclever.co/accessed on March 07, 2024
Mylab (fumo)	Wall panels	Malina Tezycka (2021)	Poland	https://fumopanels.com/accessed on March 07, 2024
FungiTech Sdn Bhd. (Boey's Mushroom and MycoPack)	Food and packaging	Boey Tze Zhou (2022)	Malaysia	https://www.fungitech.holdings/ accessed on March 07, 2024
Roha biotech	Packaging	Aditya Srinivas Kandaala (2022)	India	https://roha.bio/ accessed on March 07, 2024
Thinniyom Co., Ltd	Pet coffin (collaboration with BIOTEC-NSTDA)	Jirawan Kumsao (2023)	Thailand	https://companieshouse.co.th/cnx-healthy-products-company-limited-0505558008992 /accessed on March 07, 2024)

Table 1 (continued)

Company and (Product name™/ Commercial Product)	Biomimicry	Founders/ owner and (year founded)	Country/ Headquarter	Website/social network availability accessed
Merge Co., Ltd	Product and service: construction, art, product design, training work-shop, and business development	Ittipol Promfai (2023)	Thailand	https://www.facebook.com/profile.php?id=100063840879019/accessed on March 07, 2024)
MycoMinded	Packaging, building insulation materials, planter pots, surfboards and water filters	N/A	South Africa	https://mycominded.co.za/accessed on March 07, 2024
MycoTile	Building Materials	Mtamu Kililo and Billy Ndegwa (N/A)	Kenya	https://mycotile.eco/accessed on March 07, 2024

Remark: The selection prioritizes companies, startups, and entrepreneurs at the forefront of innovating with next-generation materials, based on sources such as Mycelium Inspired (accessed on March 07, 2024), Material Innovation (accessed on March 07, 2024), and additional web pages available from the internet in 2024. Note: Excluded from this list are direct-to-consumer enterprises that rely on external suppliers for materials rather than developing their own, internal research and development within corporations that remains confidential, startups yet to emerge from stealth mode, and any innovations in materials that do not align with the established criteria for new materials. "N/A" stands for "not applicable" or "not available."

ion" in the Netherlands [69], (Fig. 4), and a pavilion at the Rensselaer Polytechnic Institute in Troy, NY, USA [70] have utilized mycelium-based cladding panels or units atop wooden skeletons. The "MycoTree" presented at the 2017 Seoul Biennale [71], exemplified the structural use of mycelium in compression-only configurations, showcasing its potential for load-bearing applications. European research groups, such as those at Lund University, Sweden [72] and Blast Studio in London, UK [73], have developed unit-based columnar structures utilizing mycelium.

Currently, Ecovative has integrated mycelium-based panels as insulation in a small-scale housing prototype, demonstrating a practical application of mycelium in the construction industry. These projects underscore the application of myco-materials at a building scale, primarily in non-structural capacities.

Mycocrete is a construction material that includes mycelium. When infused into a fabric framework created through knitting, it generates a composite material that surpasses previous fungi-derived biomaterials in terms of durability and flexibility. This advancement may lead to the creation of buildings that are not only lighter but also have a minimal environmental footprint [74].

As shown in Fig. 4, Mogu Acoustic panels and floor tiles represent the first commercially available products of their kind composed entirely of fungal mycelium and recycled textile residues [75]. In Poland, Fumo panels [76] offer an innovative interior finishing solution, crafted from bio-based and environmentally friendly mycelium-based materials, and are biodegradable, thus posing no negative impact on the natural environment. In Portugal, MykoFoam is an insulation panel composed of renewable industrial residues, and mycelium, which provides exceptional thermal efficiency, durability, breathability, and resistance to moisture and fire [77].

Myco-space
Hy-Fi
BIOTEC-NSTDA
Mogu
The growing pavilion
Table top
Mycera
Living cocoon
Mycoschoen
MycoPack
MushLume
Mycelium chair
Sneatures
Grown.Bio
Mylo Randoseru
Helmet
Hermes-Mycowork
Mycelium paper
My™Bacon My™Jerky

Fig. 4 Examples of various commercialized mycelium products and finished prototypes are available to the public. Myco-space by V.meer (visit https://www.v-meer.de/ accessed on March 07, 2024), The Growing Pavilion by Biobased Creations (photo by Eric Melander: https://drive.google.com/drive/folders/1XdJKYEteG_RKVoUsY7WLSnqkGk5iF67F?usp=sharing; https://www.arkitectureonweb.com/en/-/projects/the-growing-pavilion), Mushlume Light (available at https://mushlumelighting.com/shop), Mogu's Pluma Panels (https://mogu.bio/pluma-panels/ accessed on March 07, 2024) and image: https://drive.google.com/file/d/15A2Xk-zKBC9GvGJtluw2FiWu_1_ryCcd/view?usp=sharing), Helmet by Nos Design and designer Diego Mata, MycoPack by Fungitech, and MyForest Foods, Myco-space (https://www.v-meer.de/my-co-space), Mycoschoen (https://elephant.art/mushrooms-somerset-house-funghi-05022020/5-kristel-peters-mycoschoen-courtesy-of-the-artist-2/, MycoPack (https://www.fungitech.holdings/about-2), Sneatures (https://germandesigngraduates.com/sneature), Hermes-Mycowork (https://www.theguardian.com/fashion/2021/jun/12/its-this-seasons-must-have-hermes-bag-and-its-made-from-fungus), Mycelium paper (https://www.nationalgeographic.com/science/article/paid-content-mushrooms-inspire-engineers), Hy-Fi (https://www.moma.org/calendar/exhibitions), Mogu (https://mogu.bio/project/deckers_milano), Living cocoon (https://loop-biotech.com/living-cocoon/), Mycera (https://www.tugraz.at/en/fakultaeten/architecture/research/research-projects/mycera-the-growing-material), MushLume (https://www.arteficius.com/brands/grown-bio), Mycelium chair (https://www.dezeen.com/2013/10/20/mycelium-chair-by-eric-klarenbeek-is-3d-printed-with-living-fungus/), Helmet (https://www.dezeen.com/2013/10/20/mycelium-chair-by-eric-klarenbeek-is-3d-printed-with-living-fungus/), Mylo Randoseru (https://tsuchiya-kaban.com/products/mylo-l-zip-purse), My™Bacon My™Jerky (https://myceliuminspired.com/the-future-of-manufacturing-introducing-mycofabrication), Mycelium paper (https://www.nationalgeographic.com/science/article/paid-content-mushrooms-inspire-engineers), Tabletop (https://www.mdpi.com/2073-4360/14/1/145)

The "Living Cocoon" developed by Dutch researchers Bob Hendrikx and Lonneke Westhoff [78] represents a sustainable and eco-friendly alternative to traditional burial methods (Fig. 4) These coffins are made from mushroom mycelium, the vegetative part of fungi, which decomposes quickly and helps enrich the soil by neutralizing toxins and providing nutrients. The decomposition process in these coffins can take as little as 30 to 45 days, significantly faster than traditional coffins, which can take years to break down. In Thailand, Thinniyom Co., Ltd. and BIOTEC-NSTDA of Thailand [79] have developed a pet coffin derived from spent mushroom substrate and corncob residues and is pre-order on sale in 2025 [80] (Supplemental Fig. 1).

Mycelium possesses natural insulating properties that make it suitable for use in refrigerators, ovens, and other appliances. Traditional synthetic insulation materials made from polystyrene and polyurethane foams are highly flammable and non-sustainable [27]. Nevertheless, for instance, insulation applications from MBC offer more significant advantages over synthetic foams and wood products and provide a major benefit in terms of fire safety compared to synthetic insulation foams. Cumulative forms of mycelium-based construction materials include block materials [81], building envelopes [82, 83], thermal insulations [84–87], particle board [23, 88, 89], and soundproofing mycelium/ acoustic materials [90, 91] (Supplemental Fig. 2).

The comparison MBCs with conventional construction materials highlights key advantages and limitations. In terms of cost-effectiveness, MBCs are more affordable than materials like steel, aluminum, and reinforced concrete, making them ideal for non-load-bearing applications such as interior panels and insulation. Regarding sustainability and recyclability, MBCs, like straw bale, cob, and rammed earth, are fully biodegradable, offering high recyclability and aligning with circular economy principles. For mechanical performance, while MBCs have lower compressive strength than concrete, steel, and bamboo composites, their strength is comparable to eco-friendly materials such as straw bale, cork, and cob. In manufacturing, MBCs are produced through low-energy fungal growth without harmful

chemical by-products, contrasting with the energy-intensive processes used for steel, glass, and bricks, as detailed in Supplemental Table 1.

Product Art Design 3D Printing / Furniture

Art, Art toy and Sculpture: Mycelium-based composites are increasingly popular in aesthetic interior design due to their unique texture. The surface of MBC products can range from smooth to rough, and their color can vary from off-white to brown, creating a natural and distinctive appearance [53]. These characteristics make MBC an attractive material for various design applications. Several categories of products have been explored, including a tabletop, a lampshade, and a bowl designed by Agata Bonenberg [53]. Various lampshade designs have been developed, such as Mycelium lights created in collaboration with Nir Meiri Studio and BIOHM, Pendant and B Wise lampshades from Myceen, and MushLume Lighting designed by Danielle Trofe and Grown.Bio [92–94], (Fig. 4). The Mycoschoen shoes, designed by artist Kristel Peterscourtesy, were showcased at the "Fungi Futures: Movements in Mycelium" exhibition organized by Somerset House in London in 2020 (Fig. 4). MYC is a disposable barbecue grill made from edible fungus, corn cob, and bamboo sticks [95]. These examples highlight the broad applicability and aesthetic appeal of MBC in interior design and everyday products.

To overcome the limitations of size and geometry in MBCs, 3D printing and digital fabrication of mycelium slurry have been explored as alternatives to traditional molding methods [72, 96, 97]. 3D printing of MBC can be achieved through liquid deposition modeling (LDM), which requires the substrate to be in a gel form before printing. Therefore, 3D-printed MBCs necessitate the inclusion of a gelling agent as an essential component. Currently, a variety of organic gelling agents, such as husk, corn starch, flour, xanthan gum, guar gum, paper cellulose, locust bean gum, and psyllium, are utilized in 3D printing formulations to achieve the necessary substrate consistency [97]. Additionally, non-digestible components such as soil, sand, and clay are incorporated into these formulations [96, 98, 99].

Notable examples of 3D-printed MBC include the Mycelium Chair created by Eric Klarenbeek, which was printed with living fungus and displayed at Dutch Design Week 2013 [100]. MyCera is a digitally printed clay-mycelium composite where the mycelium grows within the clay structure, resulting in improved mechanical properties [98, 99], (Fig. 4). Additionally, Terreform ONE used Mycofoam to fabricate seating furniture, challenging traditional manufacturing methods [101]. A significant challenge in 3D printing MBC is the development of a sterile printing system. The printing nozzle, print head, and other components must be sterilized prior to printing to prevent contamination.

Packaging

Polystyrene, in its various forms, such as foams, films, and rigid plastics, is widely used in packaging across diverse industries, including food, electronics, and device protection. Expanded polystyrene is particularly favored for its low density, durability, and insulating properties, making it an ideal packaging material. However, styrene is a petroleum-derived product, which makes it an unsustainable resource for the packaging industry. In 2012, the Ecovative Design laboratory in Green Island, New York, embarked on research to develop pilot MBC foams. Their study revealed that the properties of MBC foams were comparable to those of polystyrene foams, offering a potentially sustainable alternative to traditional

polystyrene packaging [102]. Subsequently, the company established the Mushroom® Packaging line [103], which includes MBC-based polystyrene replacements for wine shippers and gift boxes. GROWN bio, an MBC brand based in Germany and the Netherlands, produces mushroom-derived packaging for industrial applications, luxury items, and insulation [104]. Dell produces mushroom-based cushions for packaging electronic devices [105]. Established in 2020, FungiTech (Malaysia), cultivates oyster mushrooms and converts farm waste into MBC for packing purposes [106].

Commercial MBC packaging materials are, to a certain degree, compostable. GROWN bio claims their product is 100% compostable, and Ecovative Design has asserted that their packaging materials are eco-friendly and can be composted quickly. These claims are supported by the observation that a reduction of up to 60% of the total mass was observed for MBC in the ISO 20,200 soil burial test [107]. Nevertheless, the biodegradability and biocompatibility of MBC packaging materials require further investigation.

The thermo-mechanical properties of MBC foam packaging materials are comparable to expanded polystyrene foam and polyurethane foam. The tensile and compressive strengths of conventional expanded polystyrene are 7.2 and 4.8 MPa, respectively [108]. At this loading, it may elongate up to 4.9% [108]. However, when heated to 80 °C, the tensile and compressive strength values are reduced to 4.26 and 1.67 MPa, respectively [109]. Expanded polystyrene has a glass-transition temperature (*Tg*) of up to 390 °C, while polyurethane foams become amorphous at 240–255 °C [110]. The *Tg* of expanded polystyrene beads is about 350 °C [111]. Ongoing research efforts are focused on further improving the properties of MBC foams. Recent studies have shown that MBC foams derived from sawdust and rice husk exhibited approximately twice the hardness of polystyrene foams [112]. Furthermore, another study demonstrated the potential of cellulose-mycelium foams (25-day mycelium growth), which exhibit a tensile strength up to 6 MPa (with approximately 35% strain) and a *Tg* of 326 °C [113]. The study revealed that the growth of mycelium may reduce the mechanical properties of cellulose but elevate its thermal performance [113].

When comparing the toxicity of packaging materials, burning 800 g of expanded polystyrene produces twice as much nitrous dioxide and 10 times more particulate matter than burning an equivalent amount of mycelium-sawdust MBC [114]. Nevertheless, the same MBC was also reported to produce two times more sulfur dioxide and carbon monoxide during the combustion [114]. These findings highlight the need for further research into the environmental impacts of MBC, as MBCs made from different carbon sources and mycelium species could result in different by-products during their combustion.

Textiles/Fabrics/Paper-like and Fashion

Mycelium Leather: In addition to the ethical concerns of using animal products, the tanning process used to make animal skin leather requires chromium, which negatively impacts human health and the environment [115]. Leather manufactured from mycelium (mycelium leather) has properties similar to that of leather derived from animal skin and is a valid alternative material for the manufacturing of clothes and shoes. Mycelium leather can be produced in a more environment-friendly way than animal skin leather. Mycelium leather is fabricated by two main methods: liquid fermentation and solid-state fermentation [9]. Liquid fermentation often utilizes standard media or cost-effective agricultural by-products, such as blackstrap molasses, to promote fungal growth. The resultant biomass is then pro-

cessed using traditional papermaking techniques, which include fiber suspension, filtration, pressing, and drying to achieve the desired material properties [9, 116]. In the realms of industrial and construction materials, there has been a burgeoning interest in leveraging mycelium-based technologies [117]. Nonetheless, the scarcity of published experimental data, often because of proprietary knowledge or intellectual property concerns, poses significant challenges to the advancement and application of innovative mycelium-based technologies across diverse fields ranging from mycology to engineering.

Mycelium leather derived from the fully-grown cap of *Phellinus ellipsodeus* [118, 119], has been marketed as MuSkin by Grado Zero Innovation. Other examples of mycelium leather used in commercially available products include Sylvania leather (Hermes, in collaboration with MycoWork, Fig. 4), and Mylo, manufactured by Bolt Threads [45]. Mycelium leather is available in various finishes such as pebble texture, natural mycelium looks, and smoothness. Later, Fine Mycelium was developed to control the direction of mycelium thread at the microscopic level, which has superior strength, durability, and performance, and can fit a wide range of applications [120]. Mylo leather collaborated with many brands, creating the concept Yoga Mat by Lululemon, Stan Smith Mylo by Adidas, Mylo Randoseru by Tsuchiya Kaban, and Mylo Wallet Bag by objcts.io [121]. Ecovative developed AirMycelium, a high-density foam-like pure mycelium, by controlling culture environments such as air flow, nutrients, space, and humidity. AirMycelium can be compressed into a leather-like material, and its trade name is Forager [122]. Also, Forager™ developed its own vertical farm to up-scale the production of Forager in the size of 60×4 m^2 [123]. Mylea made by MYCL (Mycotech) is used in keychain, shirt, and wallet products [124]. These four majors' companies published more than 150 patents, which cover mechanical property improvement of mycelium, mycelium leather production system, and application of mycelium foam and leather [61, 125]. Apart from these established companies, several companies across the world are also focusing on the development of mycelium leather, for example, Bewider, Sporatex, NEFFA, Mylium, Mycofuture, Mycel, and Muush (Table 1).

Clothing and Shoes: Sneature is a "bio-shoe" made from natural and renewable raw materials. The shoe is made from Chiengora, a protein-based 3D knit of natural rubber and mycelium designed to enhance stiffness, flexibility, and air circulation [126]. It also requires around 20% less material in the production compared with a standard shoe. UYN URBAN TRAIL MUSHROOM MAN SHOES are also made with mycelium biomaterial protective shield layer [127]. Additionally, mycelium can be used as shoe sole because of its foam-like property [128, 129]. Mycelium shoe soles have been developed with both MBC and PMM. PMM shoe soles were developed in a similar way as mycelium leather. In 2012, Ecovative produced AirMycelium™, a dense and thick mycelium sheet. This product is related to patents describing the technology of mycelium mat/sheet [130, 131]. MBC shoe soles have also been produced in several shoe designs. Espadrille shoe soles were made with king oyster mushrooms, chicken feathers, and nonwoven textiles to test consumer opinion [128]. From this survey, it was revealed that customers would likely purchase mycelium espadrilles due to their eco-friendly and biodegradability features. A prototype sandal sole made using a combination of mushroom species and various agricultural wastes, mostly chicken feather and psyllium husk has also been reported [132, 133]. Mycelium heels made for the "Growing Shoes" project comprise solid and hollow models. While coffee grounds and sawdust were used for the heel, platform, and insole, the hollow shoe shapes are grown pure in bottles or on fibers such as hemp and psyllium to achieve strength and structures [134]. The mushroom species used for

producing MBC are under focus to identify those that confer MBC with superior mechanical properties. According to [129, 135], the MBC derived from king oyster mushrooms provides the highest compressive strength of MBC used as shoe sole. The combination of hemp fiber, leather scraps, and thermoplastic polyurethane (TPU) 3D printed filament has proven promising for shoe sole application [136]. A mycocomposite made from double layers of woven jute textile with mycelium composite inserted in the middle layer has been reported [137]. Adding laminated layers helps reinforce the mycelium composite for shoe soles [138, 139]. Lately, fire-resistant shoes fabricated from various agricultural wastes [139], and a mycelium helmet made from hay and mycelium designed to be compostable, breathable, and impact resistant [140] are examples of fire-resistant apparel that can be produced at a low cost.

Paper: Paper manufacturing causes deforestation and other negative environmental impacts resulting from the use of toxic chemicals, such as kraft, sulfite, and sodium hydroxide, in the production process. Therefore, alternative sustainable natural polymer resources to tree biopolymer for paper manufacturing are desirable. The mycelium cell wall consists of chitin, chitosan, and glucan, arranged as a complex biopolymer structure (Fig. 4). The idea of incorporating plant fiber with mycelium began in the 1970s [141]. Various fungal species cultured in liquid media were filtered and mixed with unbleached aspen kraft pulp in a ratio of 0–50% [141]. However, only *Saprolegnia ferax* could enhance the tensile strength of the paper composite. Mycelium could be used as a 5–10% paper additive without appreciable reduction of tensile strength [142]. Since chitin is one of the main components in mycelium, it can be considered a chitin/chitosan resource. Even though the mechanical properties of extracted chitin nanofiber and nanocrystal are inferior to that of nanocellulose, various studies have agreed that mycelium can produce chitin at low cost [143–146]. Chitin nanofiber can be prepared via suspension, to form a fibrous sheet suitable for paper manufacturing [146]. While the preparation by solvent casting can lead to a non-porous sheet, impurities such as alcohol and acid derivatives increase the hydrophobicity of mycelium paper. The chitin nanofiber sheet from mycelium exhibits tensile strength and moduli of more than 200 MPa and 7 GPa, respectively [143, 145].

Food Industry/and Beverage Containers

Sustainable Food Production (Protein & Meat Alternatives): Mycelium can be consumed as food in meat substitutes, gluten-free bread, and high-protein snacks. It is a meat replacement due to its high fiber and protein content, similar to that of bacon (Fig. 4). It can be grown at a competitive price of approximately $1 per pound. According to Ecovative, the greatest potential of mycelium lies in its ability to mimic whole cuts of meat. This contrasts with most other plant-based meats, which are typically offered in minced patty or sausage form [147]. As a development, MY™BACON & MY™JERKY were produced as an alternative to bacon made from AirMycelium. Meat alternative products made from fungi have existed since 1985, with Quorn® as the first protein-rich meat alternative made from *Fusarium venenatum* [148]. A higher ploidy *Fusarium* food product has been patented [149]. In Republic of Korea, *Agaricus bisporus* mycelium is another meat substitute obtainable from submerged fermentation with umami flavor characteristics, and superior texture (hardness, springiness, and chewiness) to soy protein [150].

The report of the Good Food Institute (https://gfi.org, accessed on March 07, 2024) do cumented that many new companies are betting on fermentation technologies and, more specifically, mycoproteins to further innovate in this space (e.g., ENOUGH, Mycorena,

Nature's Fynd™, The Better Meat Co., 3 F Bio, Atlast Food Co., Chunk, Fybrawork Foods, Kinoko-Tech, Meati, Mycovation, Prime Roots). In 2021, Thailand's BIOTEC-NSTDA announced Mycoprotein, a food product with a meat-like texture containing essential amino acids, fiber, vitamins, and beta-glucan, but low in fat and cholesterol-free, to be marketed as "myco-meat analogue" and "ready meal" products [151].

Nutraceuticals and Food Additives: Mycelium-based composites are used to develop foods with improved bioactive and nutritional qualities, including enhanced antioxidant and antimicrobial properties, along with increased levels of microminerals and vitamin D2. Submerged cultivation of *Pleurotus eryngii* mycelial biomass has been found to boost its nutritional benefits by increasing the content of vitamin D2 and essential minerals like calcium (Ca), cobalt (Co), and iron (Fe) [152]. In the realm of nutraceuticals, cultured mycelia-derived extracts of medicinal fungi are available as nutraceutical medicines in various forms of capsules, powders, and bottled fluids [153]. For instance, "myco-pharmaceuticals", a prescription drug in Japan for the treatment of cancer, and marketed across Asia, is composed of mushroom polysaccharide krestin (PSK) derived from *Trametes versicolor* [154–156]. Additionally, pleuromutilins are antibiotics that specifically target bacterial translation. They are semisynthetic derivatives of the naturally occurring tricyclic diterpenoid pleuromutilin, which is named after the fungus *Pleurotus mutilus*, the source of pleuromutilin [157].

The pharmaceutical company GlaxoSmithKline has also developed a pleuromutilin analog called Retapamulin, which is marketed under the name Altargo [156]. Products containing compounds like erinacines and hericenones from the *Hericium erinaceus* stimulate the release of nerve growth factor and support cultured nerve cells [158, 159]. Intriguingly, preclinical studies have indicated that incorporating *H. erinaceus* mycelium enriched with erinacines into daily meals could ameliorate health conditions such as Alzheimer's disease, Parkinson's disease, ischemic stroke, and depression. Organic ReiShi-Gen, a health supplement that combines *Lentinus edodes* and *Ganoderma lucidum*, is known for its immune-stimulating properties [160]. Regarding functional foods, research has revealed that replacing 5% of wheat flour with mushroom mycelium in fortified bread could be achieved without compromising the rheology, texture, and organoleptic properties. Specifically, substituting 5% of wheat flour with *Agaricus blazei*, *Hericium erinaceus*, *Phellinus linteus* and *Antrodia camphorate* mushroom mycelia powder results in a bread product that still retains high levels of ergothioneine (an antioxidant compound) and γ-aminobutyric acid even after baking [161].

In terms of food flavor, the distinctive taste of mushrooms is largely due to C8 volatile compounds, particularly oct-1-en-3-ol, octan-3-ol, octan-3-on, and oct-1-en-3-on [159, 162]. Flavor-enhancing compounds like guanosine-5′-monophosphate (5′-GMP), adenosine-5′-monophosphate (5′-AMP), and amino acids, particularly glutamic acid, are also present in mushrooms [163]. Furthermore, certain unsaturated fatty acids, including oleic, linoleic, linolenic, palmitic, and palmitoleic acids, as well as amino acids and nucleotides are potential flavor precursors [164]. The free amino acid contents of the mycelia of *Pleurotus eryngii*, *Agaricus bisporus*, and *Lentinula edodes* are comparable to that in their fruit bodies, constituting mycelia as a valuable raw material for free amino acid extraction [165, 166]. Moreover, the mycelia of *Cordyceps militaris* are currently being used in Taiwan for the development of nutraceuticals and functional foods [167].

MBC Applications Under Research and Development

Environmental Remediation

In the field of MBC environmental remediation, a significant focus of ongoing research is on the removal of contaminants from aquatic ecosystems. An early study demonstrated the potential of dead fungal biomass for adsorbing textile dyes from solutions. Specifically, a suspension of *Phanerochaete chrysosporium* (20 g/L) was shown to remove nearly 80% of 100 mg/L Astrazone Blue dye within an hour, highlighting the effectiveness of fungal biomass in pollutant remediation efforts [168]. A biocomposite of dead biomass of *Aspergillus niger* coated with iron oxide was employed to remove arsenic from water [169], and *A. niger mycelium* biocomposites with alginate, ferric oxide, and dopamine were successfully employed for the adsorption of various chemical species of thorium [170]. Another study used *Xylaria* to create a fungal hyphae/graphene oxide aerogel biocomposite for the removal of uranium ions [U(VI)] [171]. This biocomposite adsorbed approximately 200 mg/g of U(VI) in 30 min, with a maximum predicted adsorption capacity of 288.42 mg/g [171]. Furthermore, the adsorbent demonstrated selectivity for U(VI) even in the presence of other cations [171].

Another research showcased the development of alginate beads loaded with the filamentous fungus *Stereum hirsutum* inside nanoporous silica hydrogels [172]. The adsorbent achieved an 80% removal of malachite green within 20 h at an initial pollutant concentration of 80 µM [172]. A mycelium composite as a chemical, gas, and odor absorber has been designed for sustainable diaper and cat litter applications [173]. A Shiitake mushroom diaper was created following a similar concept [174]. A phosphate-functionalized mycelium membrane sheet was also fabricated from the dried mycelium membranes of Ecovative Design [175]. This membrane was evaluated for Pb(II) removal using a continuous cross-flow filtration system, achieving a removal efficiency of approximately 90% at a flux of 1200 LMH with a 1500 mg/L Pb(II) solution [175].

For air pollution remediation, MBC made from *Pleurotus ostreatus* was investigated for the adsorption of NO_3^- and SO_4^{2-} from atmospheric particulate matter (PM) [176]. The results showed that using hemp as a substrate could remove 204 mg/kg of NO_3^-, while a rice straw-based MBC could adsorb up to 1189 mg/kg of SO_4^{2-} [176]. The study suggested that for an MBC to adsorb PM effectively, a balance between water absorption (negative factor) and PM adsorption (positive factor) is crucial [176].

Research on soil remediation using MBCs is relatively limited, despite mycelium's natural capacity to directly absorb contaminants such as heavy metals and toxins from the soil—a process known as fungal phytoremediation [177]. Recent advancements in MBC research have facilitated the cultivation of mycelium on contaminated substrates to produce MBCs. Depending on the nature of the contamination and the fungal species employed, the toxins within the substrate can be broken down by the fungus, sequestered within the mycelium, or are accumulated in the fruiting bodies, which are then harvested for further treatment [178]. In each scenario, the myceliated substrate has the potential to be transformed into composite materials, provided that their toxicity levels are thoroughly assessed.

Agriculture, Animal-Pet, and Horticulture

Mycelium can serve as a medium in aquaponics systems to enhance sustainable food production. Mycelial networks efficiently filter water and improve the overall health of the aquaponics ecosystem by facilitating nutrient absorption and providing a habitat for beneficial microorganisms. A notable example is the work of Rhode Island School of Design students and researchers, who have created floating planting beds made of mycelium biomaterial to cleanse waterways of pollutants and restore wetland habitats. These floating Biopods return native plants to degraded wetland systems while cleansing the water through bioremediation, utilizing microorganisms that naturally decontaminate their environment. Furthermore, the biodegradable nature of mycelium ensures that the system remains environmentally friendly [179].

In addition, MBCs provide a sustainable alternative for manufacturing biodegradable gardening products such as planters, pots, and gardening tools. These mycelium-derived products are robust, lightweight, and naturally decompose over time. The use of these products reduces plastic waste and contributes to more sustainable gardening practices, minimizing environmental impact while enhancing plant growth conditions by providing a natural, nutrient-rich environment [104, 180, 181]. Moreover, mycelium-based planting blocks represent an innovative application in horticulture, offering a sustainable medium for plant cultivation. These blocks can be infused with nutrients and beneficial microorganisms to support plant growth.

As plants outgrow the blocks, the mycelium decomposes, enriching the soil with organic matter and improving soil health. This closed-loop system exemplifies sustainable horticultural practices, reducing the reliance on synthetic growing media [182]. MBCs are also well-suited for constructing beehives and pet cages, offering a natural and sustainable alternative to conventional materials. These structures leverage the insulating properties of mycelium, which helps to shelter inhabitants from extreme weather conditions. A key area of research within the HIVEOPOLIS project focuses on the modulation and sustainable integration of myco-materials into biohybrid beehive designs. This innovative approach seeks to enhance honeybee habitats by combining biological elements (such as fungal networks and bees) with artificial intelligence components (including sensors, actuators, and dancing robot filtering), ultimately improving the functionality and sustainability of future beehives [183, 184].

Health and Medical Devices and Supplies

Mycelium materials have been proposed as a tissue regenerative matrix for use in medical applications due to their unique structural properties (Fig. 5). Mycelium consists of a network of branching fibers that form the foundation of mushrooms or fungi, characterized by a microporous network. The cell wall of mycelium is composed of polysaccharides, β-glucan, and chitin, which confer significant advantages for wound healing applications by promoting epithelial regeneration and enhancing immune response efficiency [185, 186]. β-glucans can bind to receptors on immune cells such as Dectin-1, resulting in the production of cytokines and chemokines like TNF-α, IL-6, IL-12, and GM-CSF, which are crucial for immune response and inflammation [187–189]. Additionally, β-glucans within mycelium play a crucial role in promoting the maturation of macrophages, thereby enhancing their capacity

to initiate adaptive immune responses and improving the proliferation and migration of keratinocytes. This is vital for re-epithelialization, as it influences the expression of growth factors such as VEGF (vascular endothelial growth factor) and TGF-β (transforming growth factor-beta), both of which are essential for tissue regeneration and repair. Furthermore, glycosaminoglycans (GAGs) present in the outer layer of the mycelium cell wall can bind with key proteins during adhesion processes, including growth factors, cytokines, and related proteins within the extracellular matrix.

These properties highlight the high potential of mycelium for use in fabricating scaffolds for regenerative applications (Fig. 5). The gradient porous structure of mycelium closely mimics the mammalian extracellular matrix, which facilitates cell adhesion, proliferation, and the efficient transfer of nutrients and waste during cell growth, further underscoring its suitability for regenerative medicine [190]. Furthermore, myco-biomaterials show promise as smart devices for skin wound healing applications [186].

Several studies have employed chemical treatments to improve the mechanical properties of mycelium fibers for medical applications. For example, applying plasticizers to increase fiber flexibility or using chemical crosslinking techniques, such as inducing COO- functional groups via TEMPO oxidation and crosslinking with Ca^{2+}, has demonstrated shape-recovery materials [191–193]. Moreover, 3D printing was employed to fabricate mycelium hydrogel [194, 195], which was proven to influence colonization of the gel and the bridging of air gaps, thus affecting the cell growth.

A mycelium composite marketed by Ecovative has been explored for its potential as a prosthesis socket [196], which was identified based its properties of low specific gravity, low surface hardness, and acceptable thermal expansion. However, a significant drawback of mycelium composites is their extremely high-water absorption. To mitigate this issue, a hydrophobic coating would be necessary for practical application in prosthesis socket. Additionally, there is potential to develop an alternative single-layer mask that utilizes fungal mycelium as self-growing filaments, which could enhance the efficiency of individual polypropylene mask layers [197].

Amulet, Religious, and Traditional Products

The sculptures, part of the Biodegradable Sculpture Series, utilize myco-remediation as an art form, depicting human figures that symbolize a temporary existence designed to nourish the environment rather than deplete it. This work underscores a harmonious relationship with nature and advocates for responsible environmental stewardship. Drawing inspiration from themes of death, grief, decay, and regeneration, the artist delves into the cycles of life and decomposition, particularly as they relate to fungi. The artist's return to Minnesota and internship at Mississippi Mushrooms have further honed this multidisciplinary practice. The sculpture series debuted at Franconia Sculpture Park in 2021, where the pieces biodegraded over eight months as part of the Emerging Artist Residency program [198].

In 2017, Xiaojing Yan's "Lingzhi Girl" (https://yanxiaojing.com/works/lingzhi-girl; accessed on March 07, 2024) is an artwork meticulously crafted from cultivated lingzhi mushrooms, mycelium, and wood chips, exploring themes of transformation, growth, and the intersection of nature and culture. By incorporating lingzhi mushrooms, revered for their medicinal properties and cultural significance in Chinese tradition, alongside living mycelium, Yan creates a dynamic sculpture that blurs the boundaries between life and

Water purification
After
MYCOCRETE
Chemical Absorber
Cat litter
Plant pod
Floating mycelium pods
Seed tray
Mycelium scaffold
G. lucidum
P. ostreatus
2 cm
b
d
e
Sheet scaffold
ZERO POLUTION ZEBRA
Circuit board
Mycelium electrode
Foldable mycelium
Stool, NIAC
16KG
Pressure sensor
Mycelium on Martian regolith simulant

Fig. 5 Examples of innovative mycelium products in development. MYCOCRETE (https://www.iaacblog.com/programs/mycocrete-mushroom-growth-for-a-water-purification-public-deck), Floating mycelium pods (https://www.dezeen.com/2023/11/30/risd-students-floating-mycelium-pods-cleanse-waterways), Sheet sca ffold (https://www.nature.com/articles/s41598-021-91572-x), Circuit board (https://dl.acm.org/doi/10.1145/3341162.3343808), Mycelium electrode (https://www.sciencedirect.com/science/article/pii/S1566119913004497), Pressure sensor (https://www.sciencedirect.com/science/article/pii/S2369969821000670#fig0001), Mycelium on Martian regolith simulant (https://www.cnet.com/science/nasas-mars-habitats-could-be-made-from-surprising-material-fungi), Stool-NIAC (https://www.cnet.com/science/nasas-mars-habitats-could-be-made-from-surprising-material-fungi/, https://indianexpress.com/), Martian regolith simulant (https://www.frontiersin.org/journals/built-environment/articles/10.3389/fbuil.2022.965145/full), and Zero Polution Zebra (https://www.dentsu.com/sg/en/our-work/case-study-dentsu-creative-thailand-zero-pollution-zebra)

art. "Lingzhi Girl" embodies Yan's fascination with merging traditional Chinese elements with contemporary practices, highlighting concepts of regeneration and sustainability. This widely exhibited piece underscores the interconnectedness of life and emphasizes the potential for art to convey profound ecological and cultural messages [199].

In Thailand, votive figures, small statues used in religious or spiritual practices, serve as offerings to deities, spirits, or ancestors. Spirits are integral to daily life and are seen as conduits to prosperity or sources of protection. According to Thai beliefs, when a spirit grants a wish, the individual must return to the shrine and present an offering. Over the years, zebras have become popular offerings at shrines and spirit houses nationwide. To address the visual and air pollution caused by traditional votive animal figures, BIOTEC-NSTDA and the Forest Friends Project, in collaboration with Dentsu Creative Thailand, developed an innovative solution called "Zero Pollution Zebra" using agricultural waste and mycelium as the binding agent (Fig. 5). This initiative aims to reduce the environmental impact of these offerings, providing a more sustainable and eco-friendlier alternative [200]. Throughout history, divine sculptures have been made from the finest materials of their time, such as the gold-plated Buddha at Wat Pho and Michelangelo's Pietà. Furf Design Studio reinterprets these historic statues with modern materials and techniques, combining biomaterial mycelium with 3D printing technology.

This innovative method enables the creation of sustainable deities by molding mycelium in 3D-printed forms, resulting in eco-friendly sculptures that blend traditional artistic inspiration with contemporary sustainability practices [201]. The mushroom coffin, produced by the Dutch company The Loop and aptly named "The Living Coffin," symbolizes the final link in the Circle of Life, where all organisms die, decompose, and return to the natural elemental world (Fig. 4). In this process, other living organisms consume the chemical and nutrient remains of the decomposed body, completing the cycle [202]. Lastly, the Mushroom Death Suit, designed by Jae Rhim Lee and produced by Coeio, is an eco-friendly burial garment crafted from organic cotton and embroidered with thread infused with mushroom spores. As the body decomposes, the spores are activated, facilitating the breakdown of tissues and the neutralization of environmental contaminants such as heavy metals and pharmaceuticals. This sustainable alternative to traditional burial methods significantly reduces the ecological footprint by eliminating the need for harmful embalming fluids and non-biodegradable caskets. The suit promotes a harmonious return to nature, challenging conventional burial practices and encouraging environmentally responsible end-of-life choices [202].

Space and Automotive Industries

In 2020, NASA's Ames Research Center pioneered an innovative approach to space habitation through its myco-architecture project [203]. This initiative has been exploring the potential of fungi to construct living quarters for future lunar and Martian expeditions. The concept envisions a compact, MBC structure that astronauts could transport and cultivate upon arrival. The proposed habitat features a tri-layer design: an outer shell of ice for radiation shielding, a middle layer of cyanobacteria for oxygen generation, and an inner layer of mycelium that forms the actual living space. This early-stage research, supported by NASA's Innovative Advanced Concepts program, aims not only to reduce the energy costs associated with traditional habitat designs but also to explore additional applications such as water purification (Fig. 5), biomining, and self-healing structures. Brandić Lipińska et al. [204, 205] proposed using MBCs to "grow" extraterrestrial structures on-site, reducing the need for transporting heavy materials and minimizing robotic operations.

Due to their resemblance to Earth's natural elements, MBC offers various suitable properties such as self-replication, self-healing, sensing capabilities, waste processing, radiation protection, and psychological benefits. Under the mWALLd concept, Pawlicki and Bałka [206] explored the use of MBC for constructing habitats in space, particularly on Mars.

The idea emphasized methods like mold filling and 3D printing with inoculated feedstock to form large, irregular structures without the need for pre-made molds. Additionally, Hanna et al. [207] evaluated the potential of in-space manufacturing of fungal biocomposite material through preliminary experiments. Their research revealed that *Schizophyllum commune* can grow on *Azolla filiculoides* (AF) substrate under microgravity conditions, surviving for three days even at a gamma radiation dose level of 200 Gy. They produced small-scale AF/SC biocomposite panels for mechanical testing at the European Space Research and Technology Centre. They also developed tools for extrusion and 3D printing of mycelium-based pastes, creating variously shaped 3D-printed mycelium composite objects.

MBCs offer potential advantages in terms of sustainability, weight reduction, and versatility for space and automotive industries. However, further research and development are needed to overcome challenges such as scalability, durability in extreme conditions, and meeting industry standards. As technology advances, we may see increasing adoption of these materials in space and automotive applications. For example, Antolin, a global provider of technological solutions for car interiors, has developed a pioneering project that employs mycelium-based natural materials to help reduce the environmental impact of vehicle components [208].

Electronic and Informatic Device

Electronic waste, an increasing issue associated with our use of electronic devices, demands careful management. When these devices are no longer practically useful, they are often discarded, potentially leading to environmental contamination. These discarded devices contain valuable metals like gold and copper, but they also harbor hazardous substances such as lead and mercury (Fig. 5). Additionally, the manufacturing processes of conventional electronic devices and their consumables contribute to a continuous waste stream.

In the realm of electronics, MBCs and similar materials have been extensively studied as potential components of sensor devices. For instance, a study reported the electrical activity

of a slime mold placed in two blobs of agar, connected by a protoplasmic tube. This setup, within a controlled Petri dish environment, involved the use of aluminum electrodes to monitor the slime mold's electrical activity, demonstrating the potential of biological materials in electronic applications [209]. The system was used as a color sensor to differentiate red, green, white, and blue light [209]. Under the same concept of the changing power of carbohydrates, the setup was also successfully utilized for sensing different chemicals [210]. To some degree, signal transmission was also achievable with two blocks of living MBC connected by a platinum wire [211]. In addition, MBC materials have also been used as weight and pressure sensors [212, 213].

Another electronic application of MBC is their use as components of circuit boards with embedded electronic devices [214]. Danninger et al. [215] introduced the concept of MycelioTronics, which utilizes MBC to create flexible and biodegradable electronic mycelium skins. The study also showcased MBC-based batteries, which powered a sensor board to measure and transmit data. In the field of computation, it was found that living MBC materials could function as logical circuits as they transformed electrical signals non-linearly and handled complex Boolean operations [216]. However, the constant growth of living mycelium in composites caused variability in results [216]. From sensors and circuit boards to batteries and computational systems, MBC demonstrates versatility across various electronic applications. The ability to create biodegradable and flexible electronic components offers a potential pathway to reduce the environmental impact of electronic waste. While challenges remain, particularly in avoiding living mycelium for consistent performance, the progress in MBC technology represents a significant step towards more sustainable electronics and informatics.

Future of MBMs and PMMs

The future of MBMs and PMMs lies in their potential to revolutionize industries through sustainable innovation, customization, and multifunctionality [217]. These advancements offer significant opportunities to address critical environmental, social, and economic challenges while driving the transition to a bio-based and circular economy [18, 88].

Adaptive and Multifunctional Materials

The development of MBMs and PMMs is shifting toward creating adaptive and multifunctional materials that can respond to external stimuli, such as temperature, humidity, and mechanical stress. Advanced fabrication techniques, such as integrating responsive polymers or functional coatings, could enable self-healing, shape-memory, or thermally adaptive properties. For instance, adaptive bricks or panels for construction could adjust to environmental changes, enhancing energy efficiency and durability in buildings. Currently, most MBMs are designed for a single specific purpose. However, future developments are poised to transcend these limitations by focusing on multifunctionality and adaptability. By leveraging the inherent structural complexity of mycelium, these materials can be combined with advanced polymers or engineered coatings to develop self-repairing and shape-recovering capabilities [125, 218, 219]. Such innovations would allow MBMs to actively respond to various external factors, making them ideal for dynamic applications, including self-healing

bricks or adaptive concrete for construction [220]. These advancements not only enhance material performance but also support long-term sustainability by reducing maintenance and extending the lifespan of products.

Advanced Applications in Electronics and Sensory Devices

Mycelium fibers, with their intricate network structure, hold the potential to serve as communication pathways for sensors, functioning in a manner akin to neural networks. To fully harness this capability, further exploration of the electrical and optical properties of mycelium-based materials is essential, paving the way for a wide range of applications. For instance, intelligent sensing insoles have been developed as an initial step toward creating more generalized reactive fungal wearables. These insoles demonstrate how excitation wavefronts within a mycelium network can detect pressure points through electrical activity [213]. Furthermore, MBCs can be integrated into electrical circuits with nonlinear electrical properties, including applications such as printed circuit boards [221] and mycelium-based skins for sustainable electronics [215, 222].

Multi-Organism Materials/Engineered Living Composites

The development of multi-organism materials, combining mycelium with other biological organisms, is an emerging area aimed at supporting global sustainability goals. This includes integrating mycelium with seaweeds, bacterial or microbial cellulose, and plants, leveraging their cellulose, chitin, and protein structures to create biopolymer-based materials [14, 223, 224]. For instance, mycelium and bacterial cellulose can co-exist and form composites with diverse functional properties [225–227]. Examples include the creation of jute-mycelium-based vegan leather using hessian jute fabric, natural rubber, and polyhydroxyalkanoate biopolymer extracted from *Bacillus subtilis* [228], and composites made from wheat straws and polypropylene embedded with spores of *B. amyloliquefaciens* [229].

Myco-Remediation and Environmental Applications

Further research into mycelium networks and their composition is essential for optimizing their use as effective absorbers in air pollution control, such as microbial aerosol filtration [230], and in water treatment applications, where fungal pellets have shown potential for biosorbing heavy metals, dyes, phenolic compounds, humic substances, pesticides, and pharmaceuticals [231, 232]. Mycelium's role in pollutant removal includes innovative approaches like the immobilization of fungal mycelia and their enzymes, such as laccase from *Trametes hirsuta* in alginate-cellulose beads for dye decolorization [233], and the use of *Pleurotus fossulatus* for nanoparticle remediation in contaminated water, demonstrating its potential in industrial wastewater treatment [234]. Additionally, mycelium-based pellets have been effectively employed in the removal of organochlorine compounds [235], and the synergistic degradation of kerosene by non-cultivable bacterial and fungal mycelium offers another promising avenue for environmental remediation [236]. Innovations such as mycelium-based filtration membranes, bio-absorbent panels, and pollutant-sequestering composites could address critical environmental issues like microplastic contamination and toxic spills.

Thread-liked Mycelium for Woven Textile

The development of thread-like mycelium for woven textiles represents a groundbreaking advancement in sustainable material science with the potential to revolutionize the textile industry. Mycelium, with its natural hyphal network, offers a renewable and eco-friendly alternative to traditional fibers. While its current applications are mostly limited to nonwoven sheets as vegan leather substitutes, advances in cultivation and processing are paving the way for its use in thread-like structures. By optimizing growth conditions and integrating mycelium with natural polymers like bacterial cellulose, researchers can create threads with enhanced strength, flexibility, and durability, suitable for large-scale production.

Once harvested, mycelium threads can undergo mechanical or chemical treatments to improve their properties and be blended with natural fibers such as cotton or jute. These composite textiles hold promise for applications across eco-friendly fashion, medical textiles, and industrial fabrics. Mycelium's low environmental impact, biodegradability, and capacity to utilize agricultural waste make it a key material for sustainable textiles. Companies like MycoWorks, which has advanced control over mycelium fiber alignment in products like FineMycelium™, exemplify the potential of this innovation to redefine the fabric industry. The evolution of true mycelium threads could provide a high-performance, renewable alternative to synthetic fibers, addressing the urgent need for sustainable solutions in global textile production [45, 237].

Sustainable Food (mycoprotein) and Nutraceutical Production

Mycelium's rapid growth and nutrient content make it a promising candidate for sustainable food production. Future research could explore its potential in producing high-quality protein, functional foods, and nutraceuticals. Mycelium-based meat alternatives and fermented food products could address food security challenges while reducing the environmental impact of traditional agriculture. Additionally, mycelium's bioactive compounds could be harnessed for developing health-promoting supplements and medicines Mycelium, known for its ease of cultivation and rapid growth, is rich in essential nutrients such as carbohydrates and proteins, making it a promising candidate for future food production.

Its potential for mass production, whether in industrial settings or through in-house cultivation, positions mycoprotein as a novel, sustainable food ingredient with significant global market growth potential [12, 238]. Beyond technological advancements, mycelium is critical for developing food products that are not only nutritious and desirable but also scalable, affordable, and environmentally sustainable. It offers a scalable biomass solution for producing food with high-quality protein, fiber, and essential micronutrients, all crucial for improving public health. Given its environmental sustainability, mycelium-based food products are particularly well-suited to address issues of malnutrition and undernutrition, which are often exacerbated by food insecurity and diets lacking in macro- and micronutrient density [57, 239–241].

Space and Extreme Environments

Mycelium's lightweight, self-replicating, and insulating properties make it an attractive candidate for extraterrestrial applications. In space exploration, MBMs could be used to construct habitats, radiation shields, or life-support systems on Mars or the Moon. NASA's research into mycelium-based habitats demonstrates the feasibility of creating structures on-site using local resources, reducing the logistical challenges of transporting building materials [8, 9]. Similarly, mycelium-based materials could be adapted for use in extreme Earth environments, such as polar regions or deserts, where conventional materials are less effective.

Customized Mycelium Products

Advancements in biotechnology and materials science will enable the customization of MBMs and PMMs for specific applications. By tailoring growth conditions, substrate compositions, and fungal species, manufacturers can optimize properties such as strength, flexibility, thermal insulation, and water resistance. Customized products, from biodegradable packaging to high-performance insulation, will cater to diverse industries, including automotive, construction, and consumer goods.

Integration With Circular Economy Principles

As industries adopt circular economy models, MBMs and PMMs are uniquely positioned to play a central role. Their ability to upcycle agricultural waste into valuable products aligns with sustainable production practices. Future developments could focus on creating closed-loop systems where waste streams from one industry serve as raw materials for MBM production, enhancing resource efficiency and reducing waste.

Scaling and Commercialization

A critical challenge for the future of MBMs and PMMs is scaling up production to meet industrial demands. Innovations in vertical farming, automated cultivation systems, and large-scale bioreactors could improve scalability while maintaining product quality. Additionally, investments in research and development, as well as partnerships between academia and industry, will be essential to overcoming technical and economic barriers to commercialization [14].

Consumer Education and Market Acceptance

For MBMs and PMMs to gain widespread adoption, consumer education and market acceptance are crucial. Awareness campaigns highlighting the environmental and functional benefits of these materials could drive demand. Collaborations with designers, architects, and product developers will showcase the versatility of mycelium-based materials, inspiring innovative applications in sustainable fashion, furniture, and lifestyle products [15].

Challenges and Limitations in Current Synthesis Methods for Mycelium-Based Breakthroughs

Despite the growing interest in MBCs and other biocomposites as sustainable alternatives to conventional materials, several challenges persist in their synthesis methods, particularly in terms of time efficiency and scalability. These limitations hinder broader industrial adoption and commercial viability.

Extended Cultivation and Processing time

The natural growth cycle of fungal mycelium poses a significant challenge in terms of production speed. Unlike synthetic materials that can be manufactured in hours or days, MBCs and PMMs typically require several days to weeks for sufficient colonization and material formation. The duration depends on factors such as fungal species, substrate composition, and environmental conditions. Additionally, post-growth processing steps, including drying, heat treatment, and surface finishing, further extend the total production timeline.

Scalability Constraints in Controlled Growth Conditions

Large-scale production of MBCs and PMMs requires precise control of environmental factors such as temperature, humidity, and oxygen levels. Maintaining optimal growth conditions across industrial-scale production facilities is technically complex and energy-intensive. Inconsistent environmental conditions can lead to variations in material density, strength, and overall quality, posing a challenge for standardization.

Heterogeneous Material Properties

The inherent biological variability of fungal growth can result in inconsistencies in mechanical performance, density, and durability. Unlike traditional polymer-based composites, which are engineered with precise formulations, MBCs and PMMs are subject to natural variations influenced by substrate type, moisture content, and fungal metabolism. This variability makes it difficult to achieve uniform mechanical properties across large production batches.

Moisture Sensitivity and Durability Concerns

MBCs and PMMs are inherently hygroscopic, meaning they absorb moisture from the environment, which can lead to structural degradation over time. While heat or chemical treatments can improve water resistance, these processes add additional production steps, increase costs, and may introduce sustainability concerns related to chemical use.

Post-Processing and Finishing Challenges

Unlike synthetic composites that can be easily molded or machined, MBCs and PMMs often require additional surface treatments, reinforcement strategies, or composite layering to meet durability and strength requirements for specific applications. These additional

processing steps increase labor costs and production time, further complicating large-scale manufacturing.

Limited Structural Strength Compared To Conventional Materials

While MBCs and PMMs have demonstrated promising mechanical properties, their compressive and tensile strengths remain lower than those of conventional construction materials like concrete, plastics, or engineered wood. Without additional reinforcements, such as fiber additives or hybrid material integration, their load-bearing capacity remains a limiting factor for structural applications.

Energy and Resource Efficiency in Drying and Curing Processes

The drying and curing stages of MBC and PMMs production are critical for stabilizing the material and preventing unwanted microbial growth. However, these processes often require significant energy input, particularly for heat treatment methods. Achieving energy-efficient drying techniques that do not compromise material integrity remains an ongoing challenge.

Regulatory and Standardization Barriers

Due to the relatively recent emergence of MBCs in industrial applications, standardized testing protocols and regulatory approvals remain limited. Industries such as construction, automotive, and packaging require materials to meet stringent safety, fire resistance, and durability standards. The lack of universally recognized benchmarks for MBCs slows down their adoption in commercial markets.

To overcome these challenges (6.1–6.8), research efforts are focusing on optimizing fungal growth kinetics, engineering faster-growing fungal strains, and improving substrate formulations for enhanced colonization efficiency. Automation and bioreactor-based production systems may help scale up MBC and PMMs synthesis while ensuring environmental consistency. Furthermore, advancements in hybrid biocomposites—where MBCs are reinforced with natural fibers, nanocellulose, or biodegradable polymers—may enhance mechanical properties and broaden their applicability. Addressing these limitations is essential for unlocking the full potential of MBCs and biocomposites in sustainable material applications. As research progresses, further innovations in bioprocessing, material engineering, and biofabrication will likely drive the transition toward more scalable, efficient, and commercially viable production methods.

Conclusion

This review highlights the transformative potential of MBCs and PMMs in sustainable materials engineering. By utilizing agricultural residues and fungal mycelium, these biodegradable materials offer a viable alternative to synthetic options. In-depth analysis of recent patent landscaping and academic publications were revealed. Over 40 companies across five continents involved in the commercialization of mycelium-based products were systematically analyzed. Eleven key application areas have been identified, spanning commercialized

products and ongoing research in environmental remediation, healthcare, agriculture, cultural artifacts, space exploration, and electronics. These innovations address sustainability challenges while advancing multifunctional materials with self-healing, shape-memory, and thermally adaptive properties. Integrating advanced fabrication techniques with mycelium's adaptability redefines material science, enabling modular designs and multi-functional production. Prefabricated mycelium panels provide sustainable building solutions, while hybrid facilities and decentralized micro-factories enhance efficiency, reducing waste and emissions. Automation, AI-assisted cultivation, and circular manufacturing further optimize production.

Future research should focus on exploring novel fungal species, refining substrate formulations, assessing biodegradability, and improving scalability through bioreactor cultivation and AI-driven optimization. Expanding applications in healthcare, such as mycelium-based medical scaffolds and biocompatible prosthetics, could open new frontiers in biotechnology and regenerative medicine. Strengthening industry adoption will require standardization and certification to facilitate market integration and ensure consumer confidence. The adoption of MBCs and PMMs in manufacturing reduces reliance on petroleum-based materials while supporting circular economy principles. Growing consumer awareness and regulatory backing will drive market expansion, necessitating clear industry standards and certifications. Establishing interdisciplinary multisectoral collaborations between researchers, industries, and policymakers will accelerate the transition of MBCs from niche applications to widespread commercial adoption.

Overall, fungal biotechnology is reshaping sustainable material development, offering scalable, cost-effective, and environmentally responsible solutions. As MBCs and PMMs evolve, they play a crucial role in advancing a circular bioeconomy, aligning with global sustainability goals. Continued interdisciplinary multisectoral research and industry collaboration are essential to unlocking their full potential, driving material innovation for a more resilient and eco-friendly future.

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Declarations

Conflict of Interest The authors declare no conflict of interest.

Submission Declaration The work has not been previously published, is not under consideration elsewhere, has approval from all authors and relevant authorities, and, if accepted, it won't be published the same way elsewhere without copyright-holder consent.

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